Vol 12 Chapter 4 — Chemical Reactions and Thermodynamics

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 4 — Chemical Reactions and Thermodynamics

Level 1 — one sentence

Chemical reactions are substrate K-type rearrangements with thermodynamic state functions (Δ_rH, Δ_rS, Δ_rG) governing direction and equilibrium — ΔG° < 0 spontaneous; equilibrium constant $K = \exp(-\Delta G°/RT)$ — and BST views the whole process as substrate count-maximization in chemical-K-type configuration space.

Level 2 — graduate-physicist precision

4.1 Thermodynamic state functions

For a reaction $aA + bB \to cC + dD$: - Enthalpy $\Delta_r H = \sum$ products − reactants (heat at constant P) - Entropy $⍰_r S = $ entropy change (count of microstates) - Gibbs free energy $\Delta_r G = \Delta_r H - T\Delta_r S$ (driving force at constant T, P)

Spontaneous: $\Delta_r G < 0$. Equilibrium: $\Delta_r G = 0$.

4.2 Equilibrium constants

For ideal-gas reaction at constant T:

$$K_p(T) = \exp(-\Delta G°/RT)$$

with $K_p = \prod (P_i/P°)^{\nu_i}$ over products / reactants with stoichiometric coefficients ν_i.

Le Chatelier: equilibrium shifts to counteract perturbations (add reactant → forward; raise T for endothermic → forward; etc.).

4.3 Standard reduction potentials

Half-cell reactions $\mathrm{Ox} + ne^- \to \mathrm{Red}$ have standard reduction potentials $E°$ (relative to SHE = standard hydrogen electrode).

Cell potential: $E°_{\text{cell}} = E°_{\text{cathode}} - E°_{\text{anode}}$.

Nernst: $E = E° - (RT/nF)\ln Q$ — non-equilibrium concentrations.

Driving voltage of batteries, electrolysis, biochemical electron transport.

4.4 Examples

Hydrogen combustion: $H_2 + (1/2)O_2 \to H_2O$. $\Delta_r H = -286$ kJ/mol (highly exothermic). $\Delta_r G = -237$ kJ/mol. Drives fuel cells.

Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. $\Delta_r G = +2870$ kJ/mol (non-spontaneous; driven by photon energy).

Glycolysis: glucose → 2 pyruvate + 2 ATP + 2 NADH. Net $\Delta G \approx -146$ kJ/mol; couples thermodynamics to biological energy storage.

4.5 Substrate-mechanism reading

Per Casey's Principle (Vol 6 Ch 1): entropy = force = counting at depth 0. Chemical reaction directions are substrate's K-type count gradients; equilibrium = substrate count-maximum given constraints.

4.6 K-audit anchors

- Vol 6 Ch 1-4: thermodynamics foundation
- Casey's Principle (memory)

Level 3 — 5th-grader accessibility

Chemical reactions: bonds break, new bonds form. Thermodynamics tells you direction: $\Delta G = \Delta H - T\Delta S$ — if $\Delta G < 0$, reaction goes forward. Equilibrium constant $K = \exp(-\Delta G°/RT)$. Le Chatelier: perturb equilibrium, it shifts to counteract. Standard reduction potentials: drive batteries and biology. In BST: reactions are substrate K-type rearrangements; thermodynamics is substrate counting.

What comes next

Chapter 5 develops reaction kinetics and transition states.

Where to look this up

- Atkins, *Physical Chemistry*
- BST: Vol 6 Ch 1-4